higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PULP AND PAPERMAKING TECHNOLOGY
NQF LEVEL 4

13 NOVEMBER 2019

This marking guideline consists of 5 pages.

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SECTION A

QUESTION 1

1.1 B
1.2 D
1.3 D
1.4 A
1.5 B
1.6 B
1.7 C
1.8 A
1.9 D
1.10 A

(10 × 1) **[10]**

QUESTION 2

2.1 False√ – the higher the CSF, the freer the pulp (pulp drains easily)√
2.2 True
2.3 True
2.4 True
2.5 False√ – poor drying performance.√

(5 × 1) **[5]**

QUESTION 3

3.1 D
3.2 F
3.3 I
3.4 E
3.5 H
3.6 A
3.7 C
3.8 G
3.9 J
3.10 B

(10 × 1) **[10]**

TOTAL SECTION A: 25

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SECTION B

QUESTION 4

4.1
- Cleaning
- Washing
- Bleaching
- Dispersion
- Deinking (5 × 1) (5)

4.2
- Refine the pulp✓ under controlled and repeatable conditions.✓
- Form hand sheets✓ at different degrees of beating.✓
- Physically test✓ the conditioned hand sheets.✓

(markers may mark - carry out physical tests on the acclimatised hand sheets)

(3 × 2) (6)

4.3 Zero (1)

[12]

QUESTION 5

5.1 5.1.1 Phase 1: Entry
Phase 2: Sheet formation
Phase 3: Induced drainage-static
Phase 4: Vacuum drainage (4)

5.1.2 Phase 1: Entry
The stock initially impinges onto the wire or fabric.✓ The objective is to start forming a mat without sealing the sheet or causing excessive loss of fines or loading.✓

Phase 2: Sheet formation
Sheet formation starts in this stage.✓ It first starts with a low drainage in order to prevent fibre loss. There also needs to be sufficient agitation in order to prevent floc formation.✓

Phase 3: Induced drainage-static
The sheet undergoes gentle drainage by carefully selecting the correct foil blade angles.✓ Near the end of Phase 3, the resistance to drainage becomes so high that an additional vacuum is needed to continue removing water. Vacuum-assisted foil boxes or wet suction boxes are often used.✓

Phase 4: Vacuum drainage
Water removal is only achieved through the application of higher differential pressure, i.e. dry suction boxes.✓ If the sheet enters this point with too high a moisture content, there is the possibility of pin-holing, streaking and an additional loss of fines. Excessive vacuum will also cause undue wear, as well as power consumption.✓

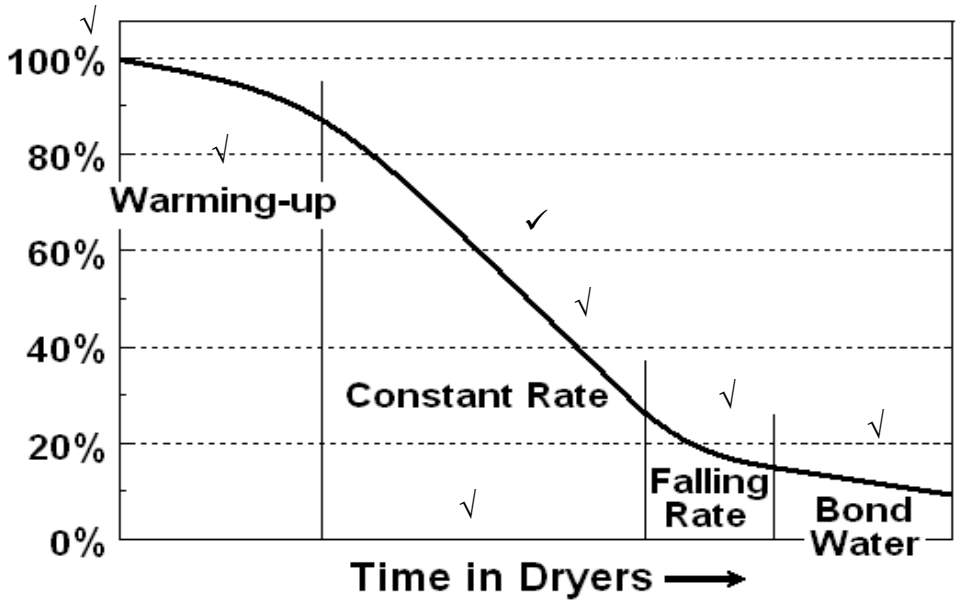
(4 × 2) (8)

[12]

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QUESTION 6

6.1 6.1.1



(ONE mark for the correct graph and THREE marks for the correct labelling) (4)

6.1.2
- The sheet will stick to the cylinders.
- The sheet will blister as steam forces the sheet apart.
- Fluff will be produced. (3 × 1) (3)

6.2
- Tension control
- Traction control
- Sheet spreading
- Web speed (4 × 1) (4)

6.3 6.3.1 Increase web tension (markers may mark - increase)

6.3.2 Reset or lubricate core chucks (markers may mark - reset or lubricate)

6.3.3 Check winder alignment

6.3.4 Increase roll tension (markers may mark - check paper making)
(4 × 1) (4)
[15]

QUESTION 7

7.1
- Black liquor is used or spent cooking liquor. (1)

7.2
- Black liquor is converted into cooking liquor,✓ which can be re-used.✓
- Black liquor releases energy,✓ which can be used to generate steam and electricity.✓
- Re-using black liquor reduces the total amount of waste✓ released into the environment.✓ (3 × 2) (6)

7.3
- The solid content of liquor needs to be at the desired concentration✓ before it can be fired into a furnace.✓
- Less energy will be needed from the evaporators if the solids content is close to the desired level.✓
- Solids content cannot be too high,✓ because it will cause equipment problems.✓
- Therefore, the quality of liquor is considered high if it has a solid content that is close to the desired solid content.✓ (3 × 2) (6)

7.4 7.4.1
A - Liquid waste feed **OR** Heavy black liquor feed
B – Exhaust gases
C – Reaction vessel **OR** Reactor
D – Cyclone separator
E – Solid product **OR** Salt cake
F – Orifice plate
G – Fluidising gas
H – Dese phase fluidised bed (8 × 1) (8)

7.4.2 A Copeland reactor produces a solid salt cake,✓ whereas a Kraft furnace produces smelt.✓ (2)

7.4.3 Primary or secondary air.✓ Fluidising air must be at a high velocity.✓ (2)
[25]

QUESTION 8

8.1 8.1.1 The balls stick to the brick and form rings. (2)

8.1.2
- Drying the lime mud
- Raising the temperature
- Maintaining a high temperature (3 × 1) (3)

8.2
- Scaling deposits are huge
- Raw materials are inexpensive
- Risk of disposing effluent into the sea (Any 2 × 1) (2)

8.3
- Dry matter content
- Available lime
- Acid-soluble sodium
- Water-soluble sodium (4 × 1) (4)
[11]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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